

Multi-Modal, Training-Free Crack Extraction via Generalized Frangi Graph

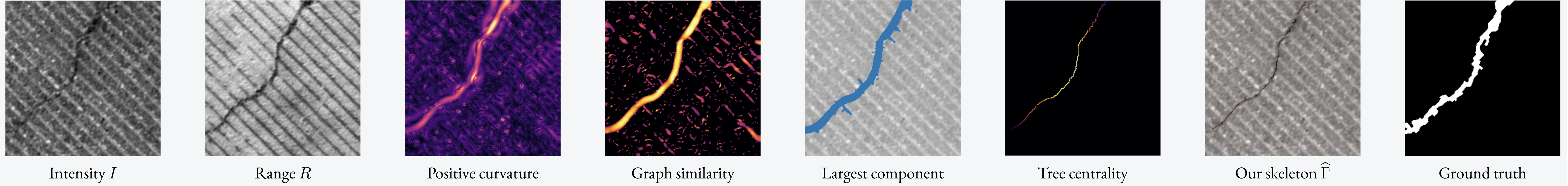
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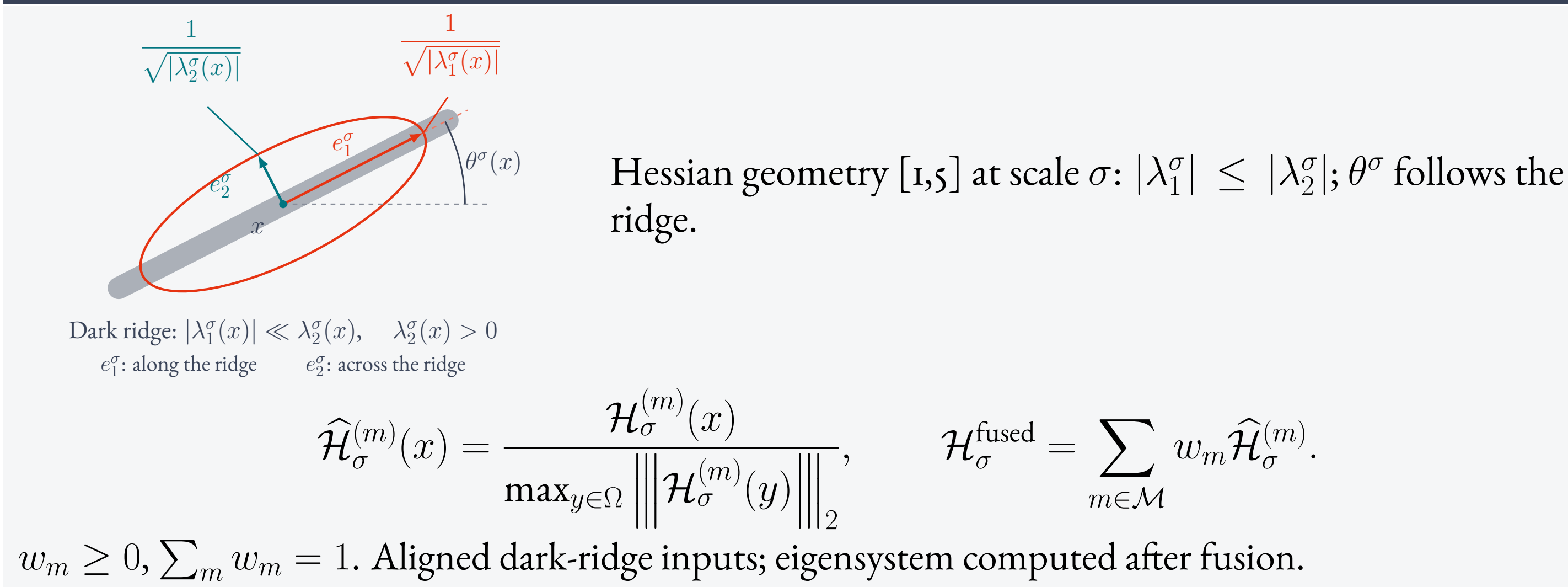
²Cerema, END SUM research team, France

From aligned images to a crack skeleton: without training

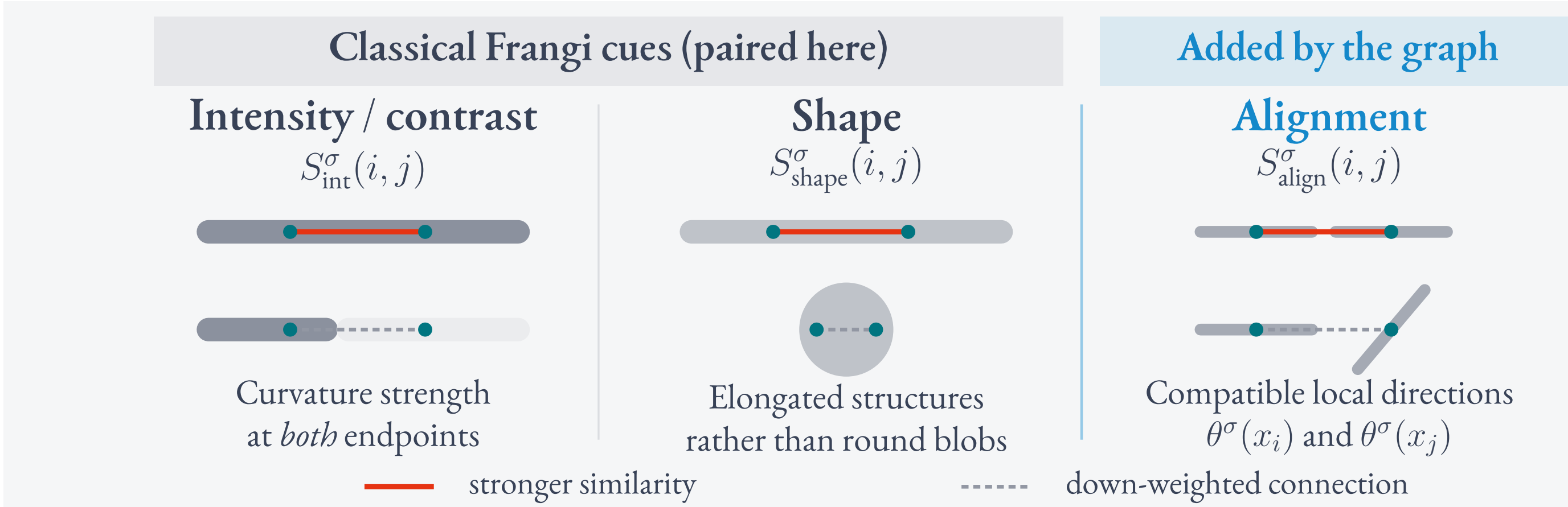
Fuse geometry, connect compatible pixels, extract a centerline. No training labels required.



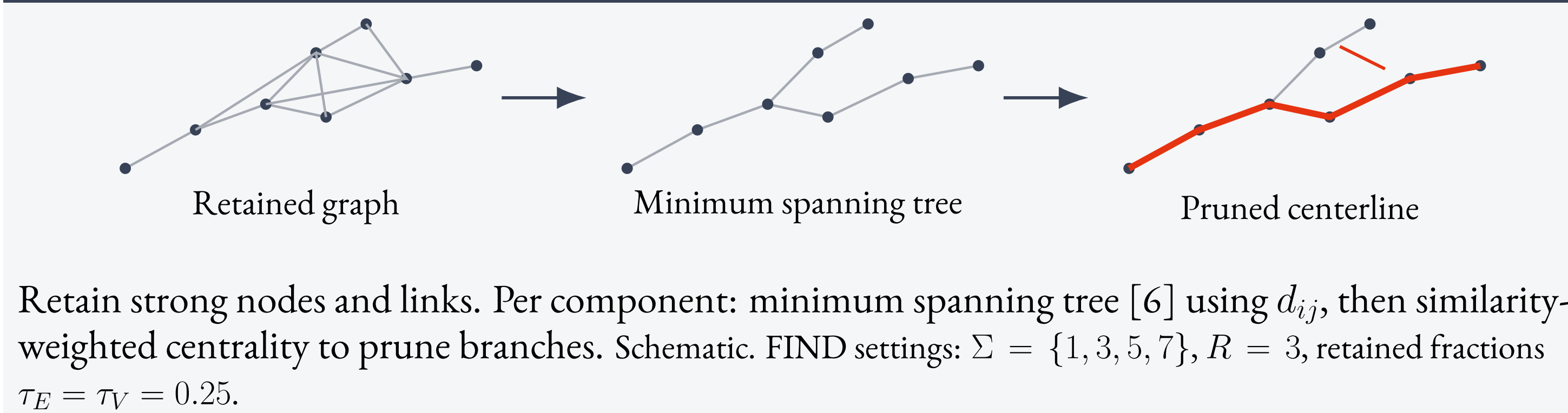
1. Fuse the geometry across modalities



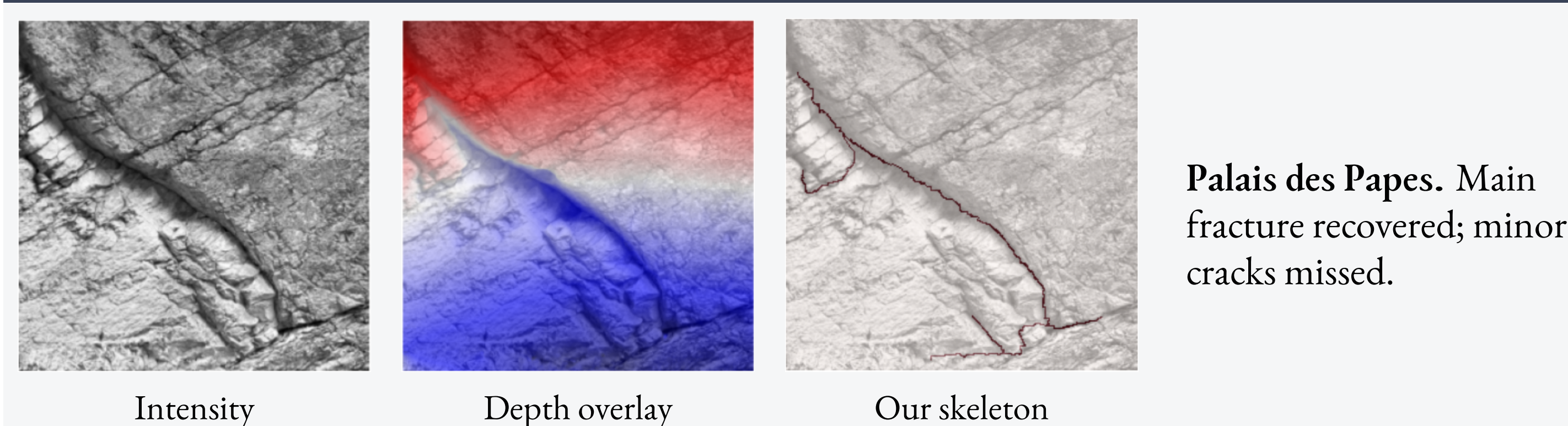
2. From pixel-wise Frangi [1] to pairwise links



3. Reduce the graph to its main centerline



A second setting: a fractured limestone wall



4. FIND: clean accuracy and noise sensitivity

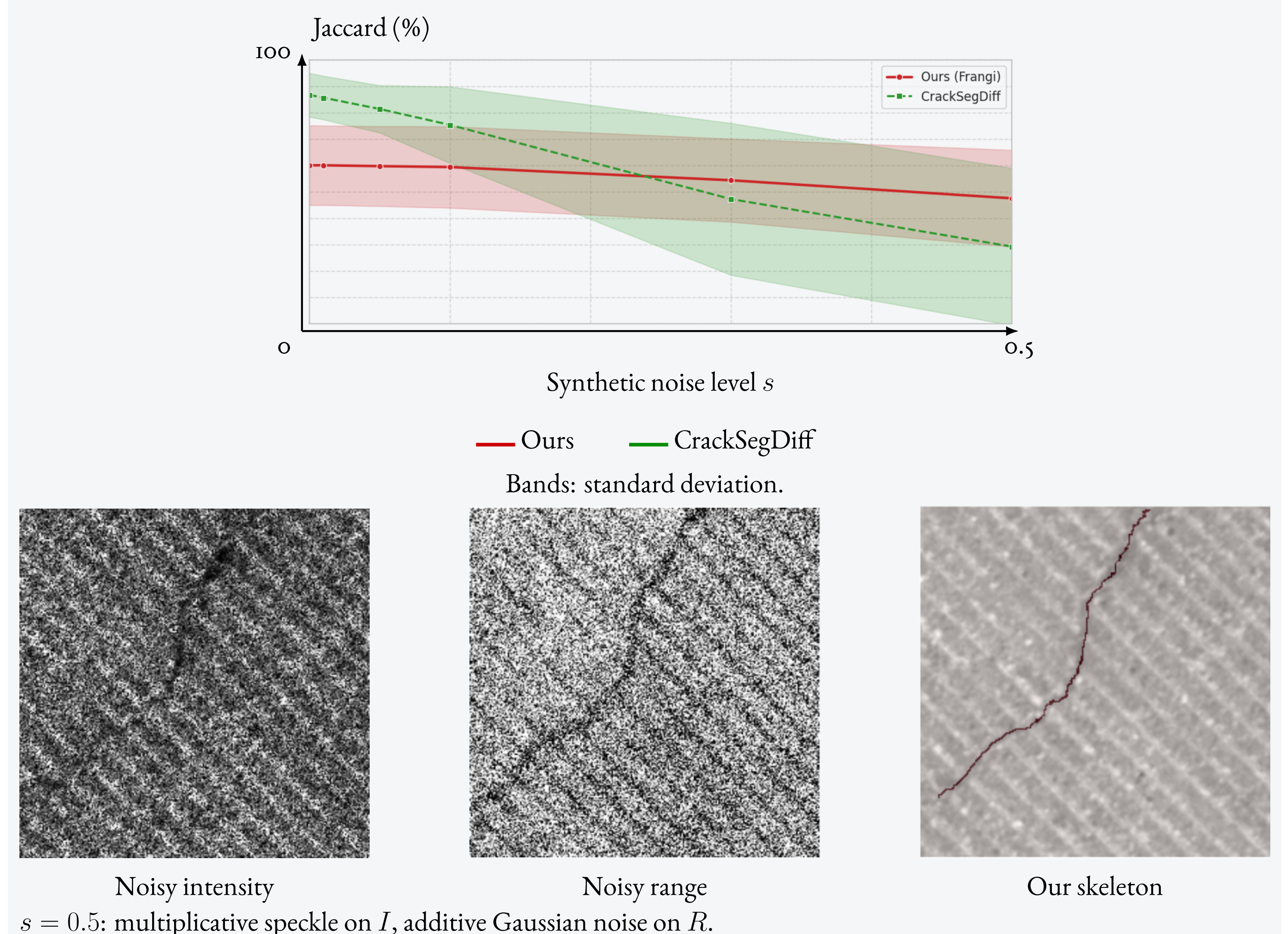
Clean FIND: the supervised model performs best.

Method ($I + R$)	Jaccard ↑	Tversky ↑	W ↓
Ours, no training	60%	68%	18 px
CrackSegDiff [3]	87%	90%	5 px

FIND [2], 256 × 256. Overlap: 3 px dilation of both skeletons; Tversky: $\alpha = 1, \beta = 0.5$. W : skeleton Wasserstein distance.

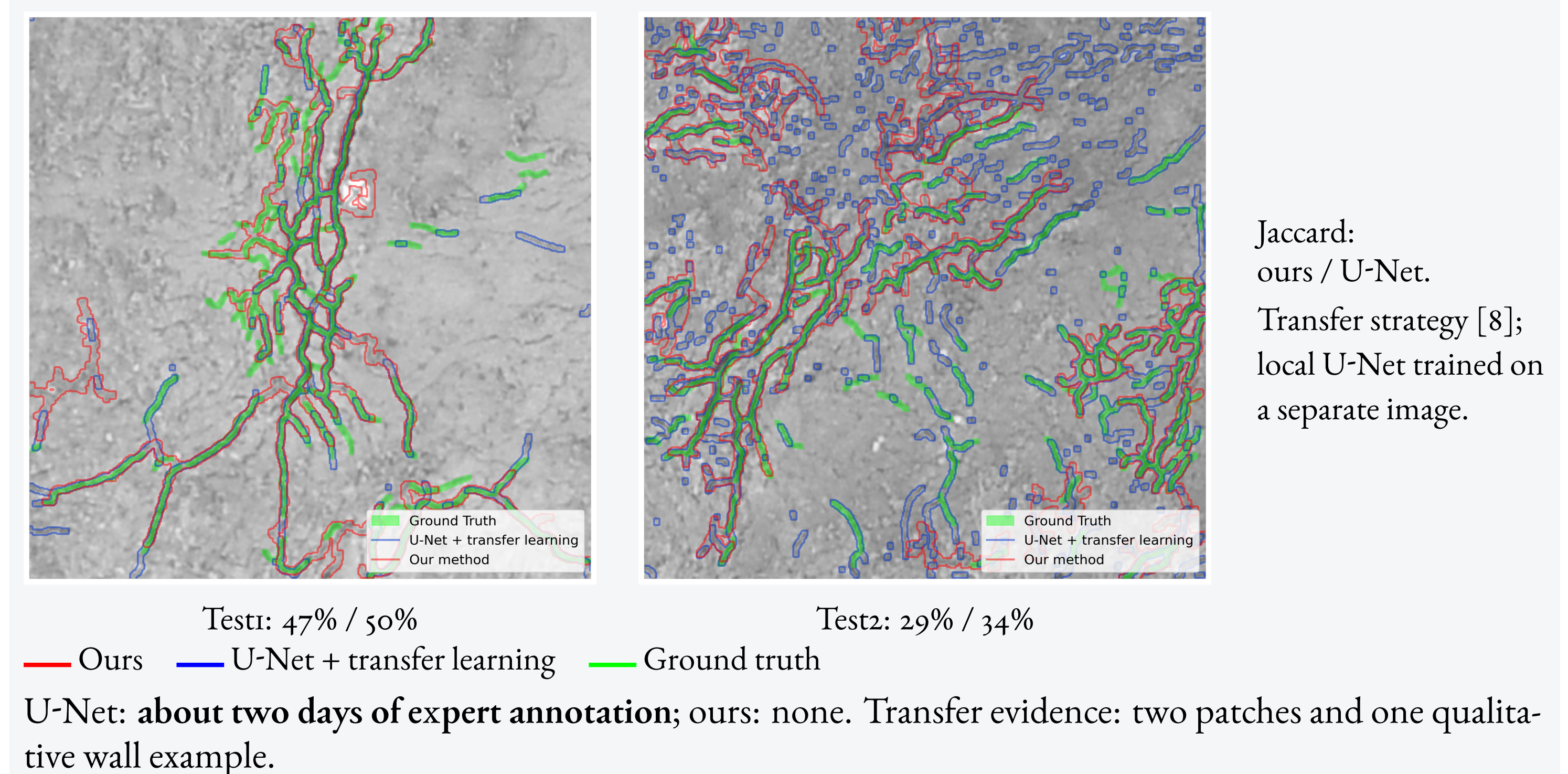
Fusion improves Jaccard: I : 41%; $I + R$: 60%; $I + R + F$: 63%. Equal weights; F is filtered range.

Slower degradation under the tested synthetic noise.

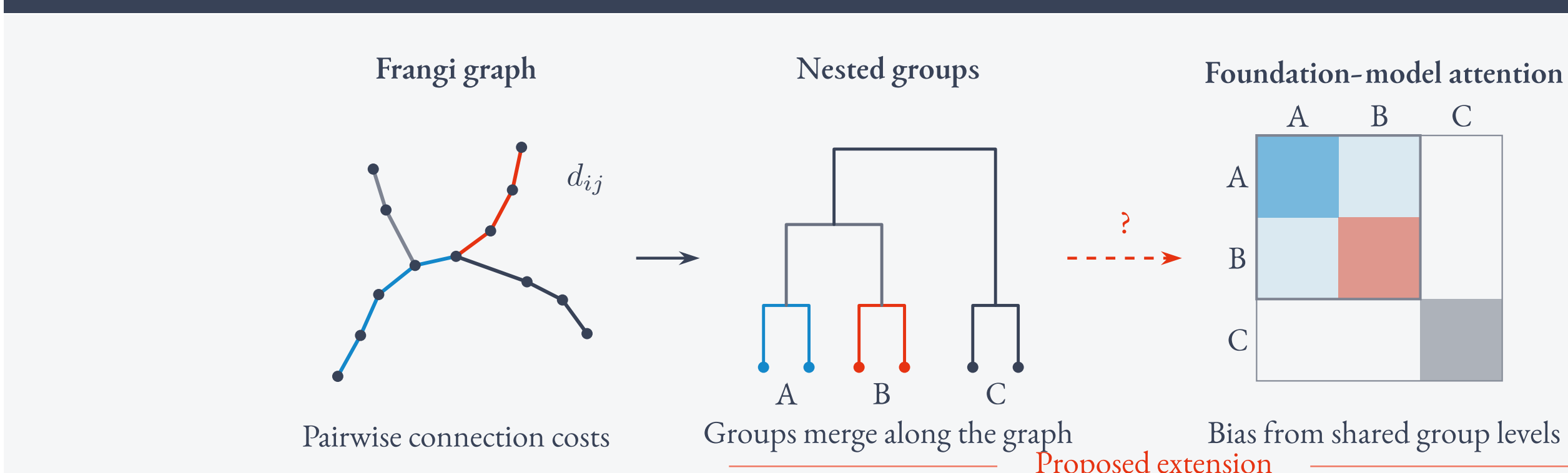


5. Geological transfer without training labels

Vaches Noires. Two Cerema UAV patches (512 × 512).



Outlook: could a Frangi hierarchy guide a foundation model?



Could nested Frangi groups guide SAM [7] / CrackSAM [4] attention, from local segments to the whole network?

Hypothesis: not evaluated here.

References

[1] Frangi et al., *Multiscale vessel enhancement filtering*, MICCAI, 1998.
 [2] Zhou et al., *FIND*, Zenodo, 2022.
 [3] Jiang et al., *CrackSegDiff*, arXiv:2410.08100, 2024.
 [4] Ge et al., *Fine-tuning vision foundation model for crack segmentation in civil infrastructures*, Constr. Build. Mater. 431, 136573, 2024.
 louis.hauseux@inria.fr · github.com/Ayana-Inria/Frangi-EUVIP

[5] Sato et al., *Three-dimensional multi-scale line filter for segmentation and visualization of curvilinear structures in medical images*, Med. Image Anal. 2(2), 1998.
 [6] Prim, *Shortest connection networks and some generalizations*, Bell Syst. Tech. J. 36(6), 1957.
 [7] Kirillov et al., *Segment Anything*, ICCV, 2023.
 [8] Shi et al., *Semi-universal geo-crack detection by machine learning*, Front. Earth Sci. 11, 1073211, 2023.



Code & data